YAKEEN

NEET 2.0 2027

Physics | Lecture - 06

Motion in a Straight Line

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Todays Goal

PhD on acceleration

H/W goal

Sangharsh – 2 (1-D)

MR. Physics

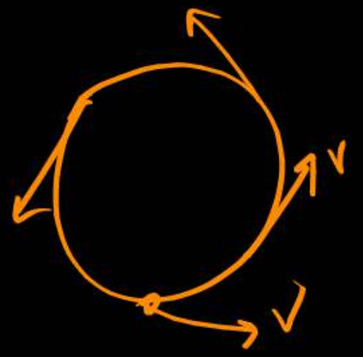
1-D Question

Question no mention

(Q) To change direction of motion object must comes to at rest once.

(a) True (-1)

(b) false (+1)



MR* Boy

In 1-D motion (straight line)
To change dirn (v=0) at once

Acceleration → ye motion (How fast) ka feel nahi hai

→ change in velocity per sec ka feel hai

→ diretion along change in velocity (or force)

$$\vec{a}_{Avg} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$$

$$\vec{a}_{gnst} = \frac{d\vec{v}}{dt} = \frac{d^2(x)}{dt^2} = v\left[\frac{dv}{dx}\right]$$

* $\langle a \rangle = \frac{\int a\,dt}{\int dt}$

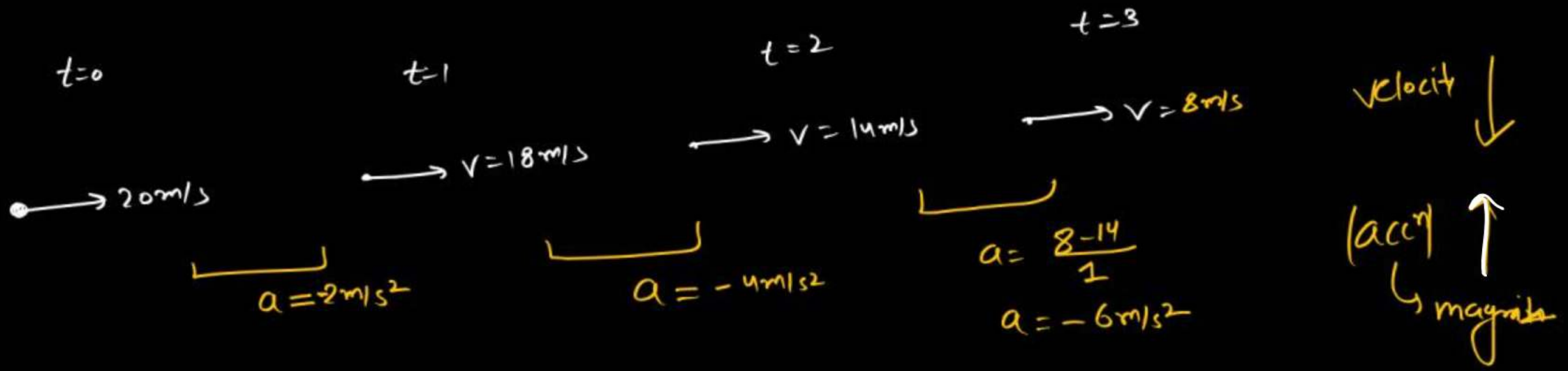
→ m/s²

→ vector

→ Rate of change in velocity (How fast/where)

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time t=0	t = 1s	t = 2s	t = 3s	acceleration	
① → 10m/s	→ V=10m/s	→ V=10m/s	→ V=10m/s	$\vec{a} = \frac{\Delta v}{\Delta t} = 0$	zero accⁿ ∴ Constⁿ velocity (uniform motion)
② → 10m/s	$a = \frac{20-10}{1-0} = 10m/s^2$ → V=20m/s	$a = \frac{30-20}{2-1} = 10m/s^2$ → 30m/s	$a = \frac{40-30}{3-2} = 10m/s^2$ → 40m/s	$\vec{a} = 10m/s^2$ (uniform accⁿ)	Cotⁿ accⁿ; Velocity ↑ (Increasing) (Non uniform velocity) — Non uniform Motion
V=10m/s	$a = 10m/s^2$ → 20m/s	$a = \frac{28-20}{1} = 8m/s^2$ → V=28m/s	$a = \frac{34-28}{3-2} = 6m/s^2$ → V=34m/s	⇒ accⁿ is continously (decreasing) (Non-uniform accⁿ) ⇒ velocity is continously ↑ (Increasing)	
→ V=10m/s	$a = 10m/s^2$ → V=20m/s	$a = 15m/s^2$ → V=35m/s	$a = 20m/s^2$ → V=55m/s	accⁿ is Contin ↑ (Both accⁿ & velocity Non-uniform) Velocity is increasing ↑	
→ V=20m/s	→ V=10m/s	→ V=6m/s	→ V=4m/s²	velocity (↓) accⁿ (↓)	



at any inst. $v = 0$ but $a \neq 0$

$v = 0$: Motion rest

$a \neq 0$: velocity me change

$\Rightarrow$ Possible at max^m height of motion under gravity

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Question

No Copy

An object is moving with constant velocity then which of the following option is correct

1. Acceleration may be increasing.
2. Acceleration is zero.
3. Acceleration is decreasing.
4. Acceleration is non-zero.

① which of the following option is correct for velocity & acc^n

magnitude of acc^n

(a) $v\uparrow$; $a\downarrow$ ⟹ Possible(✓)

(b) $v\uparrow$; $a = \text{cost}^n$ ⟹ ✓

(c) $v\downarrow$; $a\downarrow$ ⟹ ✓

(d) $v\uparrow$; $a\uparrow$ ⟹ ✓

(e) $v\downarrow$; $a\uparrow$ ⟹ ✓ (magnitude of acc^n)

(f) $v = \text{cost}^n$; $a \neq 0$ ✗ wrong

(g) $v \neq 0$; $a = 0$ Uniform moti ✓

(h) $v = 0$; $a \neq 0$ Possible at max^m height in moth unth gravity ✓

($v = 0$ rest)

$a = \frac{dv}{dt}$ ✓

3. Object is moving with acceleration $2\ m/s^2$ its velocity at $t=0$ is $10\ m/s$ then find its velocity at $t = 4\ sec$.

Costly

$+8$

$V_{f(t=4)} = 18\ m/s$ ✓

$V = u + at = 10 + 2 \times 4 = 18\ m/s$ ✓

2. velocity at $t=0$ sec is $10\ m/s$ its velocity becomes $40\ m/s$ after 6 sec then find acceleration.

$a = 5\ m/s^2 = \frac{\Delta V}{\Delta t}$ ✓

4. velocity at $t = 2$ sec is $20\ m/s$ at $t = 5$ sec it becomes $32\ m/s$ then velocity at 7 sec. will be

$a = \frac{V_f - V_i}{\Delta t} = \frac{32-20}{5-2} = \frac{12}{3} = 4\ m/s^2$

$a = 4\ m/s^2$

$V = 40\ m/s$

$\rightarrow V = u + at = 20 + 4 \times 5 = 40\ m/s$

(5.) Some wrong example :-

(a) ✗ Velocity Increasing but $acc = 0$

(b) ✗ $v = cost^n$ but $acc^n \neq 0$

(c) ✗ $v = cost^n$ and $a \downarrow$

(d) ✗ $v = cost^n$ & acc^n non zero constant

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Question

6

$\vec{u}_i = 40\hat{j}$

$\vec{v}_f = 40\hat{i}$

Kallu is moving with speed 40 m/s in north after 10 sec he is moving with 40 m/s in east then find

(i) Magnitude of rate of change in velocity. (acc^n)

(ii) Rate of change in magnitude of velocity. (speed)

$$\vec{a} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t} = \frac{40\hat{i} - 40\hat{j}}{10-0}$$

$$\vec{a} = 4\hat{i} - 4\hat{j}$$ ✓

$$|a| = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

$|v_f| = 40$

$|v_i| = 40$

change in speed is zero.

Question

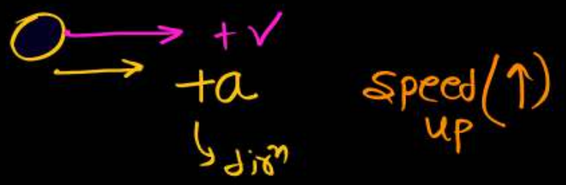
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Which of the following option is correct:

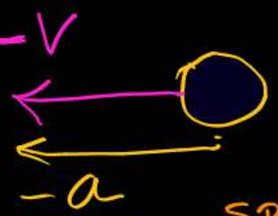
1. Velocity of object increasing and acceleration may decreasing.
2. Velocity of object decreasing and acceleration may increasing.
3. Acceleration may be non-zero when velocity of object is zero.
4. All of these.

meaning of +ve & -ve acceleration

+ve & -ve represent direction of acc^{n}.

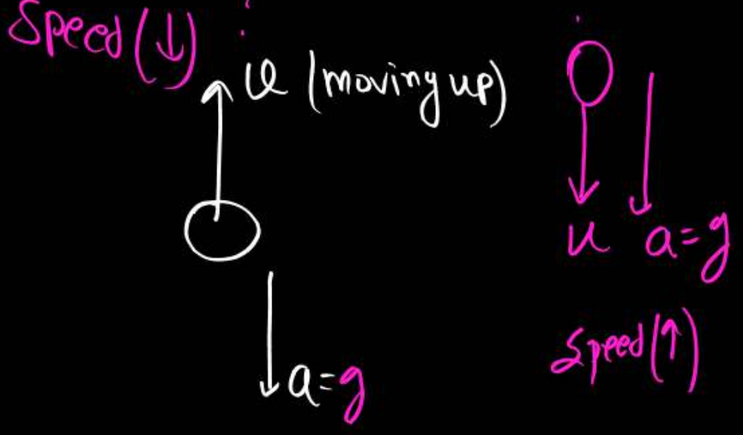


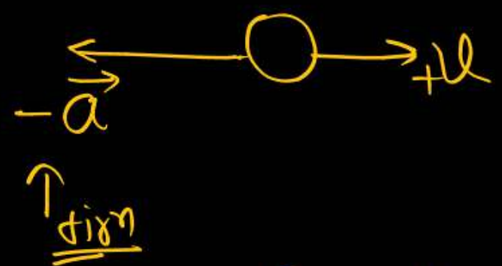
speed (↑) up



speed (up) ↑



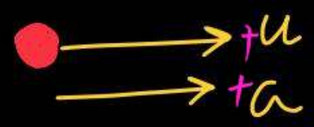




• Spee down. (↓)



Speed (down) ↓



$\vec{V}$ & $\vec{a}$ are Parallel

$\theta = 0^\circ$

Speed ↑

⇒ Motion in 1-D without change in direction

⇒ Non-Uniform motⁿ



- $\vec{u}$ & $\vec{a}$ are Anti-Parall.
  $\theta = 180$
- Speed ↓

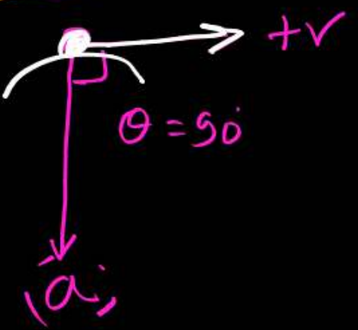
- Motion with decreasing speed comes to at rest after some time & take U-turn & move backward;

Non-Uniform motion

1-D motion

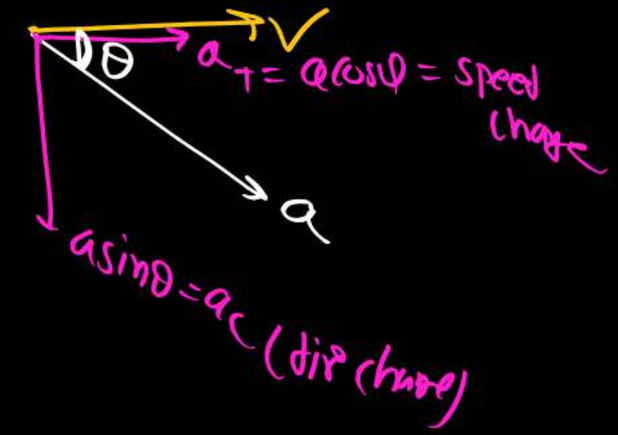
Tangential accⁿ → acⁿ which is along Path of object

2-D motion



$\vec{a}$ is ⊥ᵣ $\vec{V}$

at this instant dirⁿ will change this acⁿ is called normal or centripetal accⁿ ✓



Question

Object is moving such that its velocity and acceleration is in opposite direction then

1. Speed may constant.
2. Speed may increasing.
3. Speed must be decreasing.
4. Speed may be increasing or decreasing.

MR* Box :-
object ka future path velocity & acceleration ke bich ka Angle decide karega.

MR* Box :-
If velocity & acc^{n} Parallel speed up

velocity & acc^{n} anti-Parallel speed ↓

Meaning of +ve and −ve acceleration :−

+ve and -ve only depends on direction.

Retardation ?

(Slow down)

⟶ -ve acc^n does not mean retardation ✓

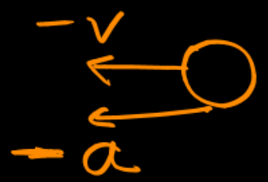
⟶ acc^n which is opposite (Anti-Parallel) to the direction of velocity c/d retardation.



Speed ↓ (retn) ✓

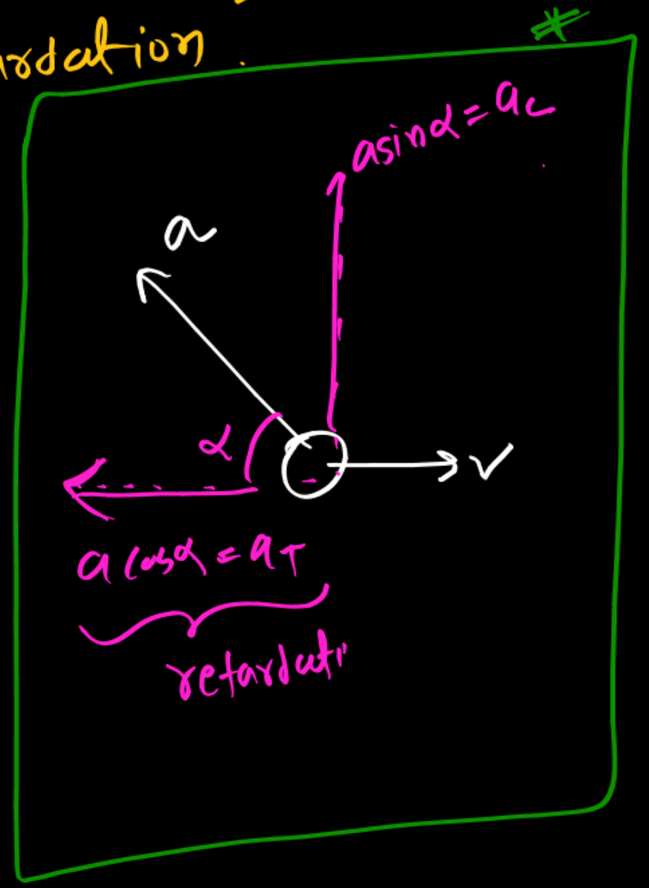


✓ retardation (speed ↓)



Speed ↑ (Not retardn)

✗



$a = 10\ m/s^2$

$a = 12\ m/s^2$

$a = 5\ m/s^2$

Ex → $a = 2\ m/s^2$

variable acc.

$a = 5t^2$

$a = \sin(t)$

$a = kx$

$acc^n = 0$	$acc^n = cost^n$ (uniform acc^n)	acc^n = variable.
→ $\vec{u} = cost^n$	→ velocity varible (Non-uniform motn with cut^n acc^n)	↓
uniform motn	• apply equation of motn.	can't apply eqn of motn ✓
$s = ux t$	$\vec{v} = \vec{u} + \vec{a}t$	use integration
Path 1-D without change in dirn.	$\vec{s} = \vec{u}t + \frac{1}{2}at^2$	⇒ diff^n
	$v^2 - u^2 = 2as$	$a = \frac{dv}{dt}$ ✓

Vali Concept

Displacement $\Delta x = \int_{x_i}^{x_f} dx = \int v\,dt$ = Area of v/t graph

Change in velocity $\Delta V = \int_{u}^{f} dv = \int_{ti}^{tf} a\,dt$ = Area of a/t graph

वालि = $\int x\,dt$

Area of x/t graph → वालि

Position $\vec{x}$ ⇄ $\vec{V}$ ⇄ $\vec{a}$

$\vec{V}$ → वालि

diff^n $\vec{V} = \frac{d\vec{x}$

velocity = Slope of x/t graph

diff^n of v(t) $\vec{a} = \frac{d\vec{v}}{dt}$

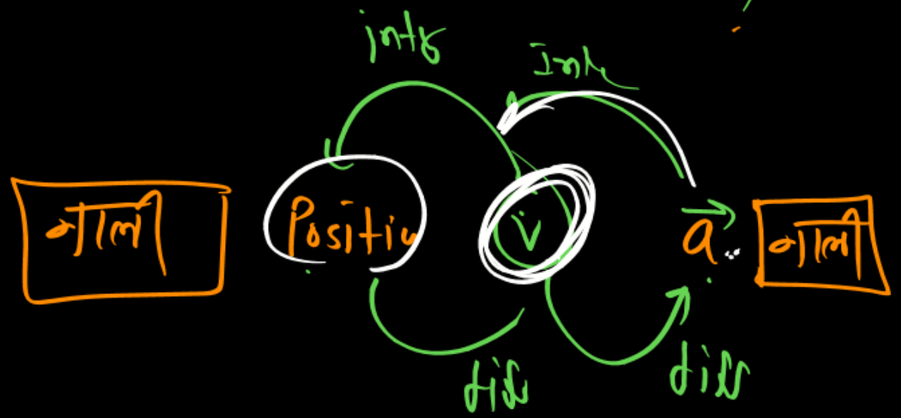
acc^n = Slope of v/t is $\left(\frac{dv}{dt}\right)$

gf v(x) $\vec{a} = v\frac{dv}{dx}$

$\vec{a}$ → वालि

$\frac{d\vec{a}}{dt}$ = Jerk (Not in Syllabus)

Slope of a/t graph = वालि



acceleration = 0	acceleration = cost^{n}	acc^{n} = variable

find acceleration in each question

$x = 4t^2 + 6$

$v = \frac{dx}{dt} = 4(1) + 0$

$v = 4$

$a = \frac{dv}{dt} = 0$

$x = 3t^2 + 4(t) + 6$

$v = 3[2t] + 4(1) + 0$

$v = 6t^1 + 4$

$a = 6[1] + 0 = 6 m/s^2$

$a = const$

$x = 2t^3 + 5t$

$v = 2(3t^2) + 5$

$v = 6t^2 + 5$

$a = 6(2t) + 0$

$a = 12t$

variable acc^n

$x = t^4 + 4t$

$v = 4t^3 + 4$

$a = 4(3t^2)$

$a = 12t^2$

acc^n = variable

$v = 3t^2 + 4$

$a = 3(2t) + 0$

$a = 6t$

$v = 3t + 4$

$a = 3(1) + 0$

$a = 3$

const

$v = t^3 + 4$

$a = 3t^2$

$v = 2x + 4$

diff^n w.r.t x

$\frac{dv}{dx} = 2x\frac{dx}{dx} + \frac{dy}{dx}$

$= 2 \times 1 + 0$

$= 2$

$a = v\frac{dv}{dx} = (2x+4) \times 2$

variable acc^n

If $x = \sqrt{t}$ then find acc^n

$$x = t^{1/2}$$

$$V = \frac{dx}{dt} = \frac{d\,t^{1/2}}{dt}$$

$$V = \frac{1}{2} t^{\frac{1}{2}-1} = \frac{1}{2} t^{-1/2}$$

$$a = \frac{dv}{dt} = \frac{1}{2}\left(-\frac{1}{2}\right) t^{-\frac{1}{2}-1}$$

$$\# \quad a = -\frac{1}{4} t^{-3/2}$$

If $x = \frac{1}{\sqrt{t}} \propto t^{-1/2}$

↳ a (variable)

② If $v = k\sqrt{x}$ then find acc^n

Soln diff^n w.r.t x

$$\Rightarrow \frac{dv}{dx} = k\frac{dx^{1/2}}{dx}$$

$$= k\frac{1}{2}x^{\frac{1}{2}-1}$$

$$\Rightarrow \left(\frac{dv}{dx}\right) = \frac{k}{2}x^{-1/2}$$

$$a = v\left(\frac{dv}{dx}\right) = k\sqrt{x}\cdot\frac{k}{2}x^{-1/2}$$

$$a = \frac{k^2}{2}\frac{\sqrt{x}}{\sqrt{x}} = \frac{k^2}{2} \Leftarrow \text{Const}$$

$v = kx$

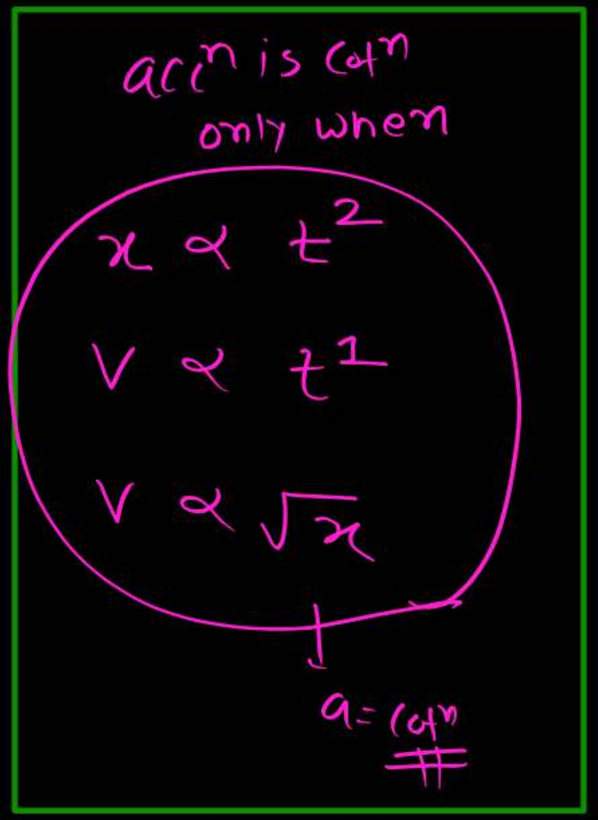
$v = kx^2$

$v = \frac{1}{\sqrt{x}}$

$v = x^{-2}$

$v = kx^3$

acc^n → variable


acc^n is c^{ot} only when
$x \propto t^2$
$v \propto t^1$
$v \propto \sqrt{x}$
a= c^{ot}

Velocity of object $v = 2x^3$ then find acceleration ??

gm Which case acceleration is uniform (constant)

(1) $x = \sqrt{t} + 4$ ✗

(2) $\sqrt{x} = t - 3$ ✓

(3) $x = t^{-2} + 4$ ✗

(4) $x = \frac{4}{t}$ ✗

(5) $x = t^3 + 4$ ✗

(6) $x = \frac{3}{t^2}$ ✗

(7) $x = 4t^2 + 6$ ✓

(8) $v = 3t^2 + 4$ ✗

(9) $v = 6t - 4$ ✓

(10) $v = \sqrt{x}$ ✓

(11) $v = x^2 + 4$ ✗

(12) $v = 2x + 3$ ✗

(13) $a = 2t$ ✗

(14) $a = 3x$ ✗

(15) $a = 2t^2$ ✗

(16) $a = 5 m/s^2$ ✓

(17) $a = 3\sqrt{x}$ ✗

(18) $x = A\sin(\omega t)$ ✗

(19) $v = A\cos(\omega t)$ ✗

(20) $a = 2\sin(t)$ ✗

Question

If velocity $V = k\sqrt{x}$ then which of the following is correct for position [IIT-2006]

1. $x \propto \sqrt{t}$
2. $x \propto t^3$
3. $x \propto t^{-2}$
4. $x \propto t^2$

Question

If the displacement of a particle varies with time as $\sqrt{x} = t + 7$, then

1. Velocity of the particle is inversely proportional t
2. Velocity of the particle is proportional to t^2
3. Velocity of the particle is proportional to $\sqrt{t}$
4. The particle moves with constant acceleration

$x = (t+7)^2$

$a = \text{const}$

$v = ?$

Question

If position $x = t^2 + 5t^3 + 6$ then find

(i) Initial acceleration.

(ii) Initial velocity.

(iii) Acceleration at $t = 2$ sec.

Question

If position $x = at^2 - bt^3$ find the acceleration is zero.

(Q) Position of object $x = t^3 - 6t^2 + 10$ find time when acceleration of object will be zero.

Position of object $x = 2t^3 - 4t^2 + 4$

find velocity, acceleration and Position

at $t = 2$ Sec

Position $x = t^3 - 9t^2 + 4$. find velocity when acceleration is zero.

Position of object $x = 2t^3 - 24t + 4$ find acceleration when velocity of object is zero.

Time is given as $t = x^2 + x$ then acceleration of the Particle as a function of Position.

Question

The position x of particle moving along x-axis varies with time t as $x = A \sin(\omega t)$ where A and w are 127 positive constants. The acceleration a of particle varies with its position (x) as

1. $a = Ax$
2. $a = -\omega^2 x$
3. $a = A\,\omega\,x$
4. $a = \omega^2 x\,A$

Question

The initial velocity of a particle is u (at $t = 0$) and the acceleration a is given by $\alpha t^{3/2}$. Which of the following relations is valid?

1. $v = u + \alpha t^{3/2}$
2. $v = u + \frac{3\alpha t^3}{2}$
3. $v = u + \frac{2}{5}\alpha t^{5/2}$
4. $v = u + \alpha t^{5/2}$

Question

A particle moves along a straight line such that its displacement at any time t is given by $s = (t^3 - 6t^2 - 3t + 4)$ meters. The velocity when the acceleration is zero is

1. 3 m/s
2. 42 m/s
3. – 9 m/s
4. – 15 m/s

Question

Position of object $x = t^3 - 6t^2 + 10$. Find time when acceleration of object will be zero.

Question

Position of object $x = 2t^3 - 4t^2 + 4$. Then, find velocity, acceleration and position at $t = 2$ sec?

Question

The distance travelled by a particle is related to time t as $x = 4t^2$. The velocity of the particle at $t = 5$s is: **[25 Jan, 2023]**

1. $40\ ms^{-1}$
2. $25\ ms^{-1}$
3. $20\ ms^{-1}$
4. $8\ ms^{-1}$

Question

The position of a particle related to time is given by $x = (5t^2 - 4t + 5)$m. The magnitude of velocity of the particle at $t = 2$s will be: **[15 April, 2023]**

1. $10\ ms^{-1}$
2. $14\ ms^{-1}$
3. $16\ ms^{-1}$
4. $06\ ms^{-1}$

Question

The distance travelled by an object in time t is given by $s = (2.5)t^2$. The instantaneous speed of the object at $t = 5$s will be: **[13 April, 2023]**

1. $12.5\ ms^{-1}$
2. $62.5\ ms^{-1}$
3. $5\ ms^{-1}$
4. $25\ ms^{-1}$

Question

The velocity of a particle is $v = v_0 + gt + Ft^2$. Its position is $x = 0$ at $t = 0$; then its displacement after time ($t = 1$) is: **[17 March, 2021]**

1. $v_0 + g + f$
2. $v_0 + \frac{g}{2} + \frac{F}{3}$
3. $v_0 + 2g + 3F$
4. $v_0 + \frac{g}{2} + F$

Question

The position of a particle as a function of time ts, is given by $x(t) = at + bt^2 - ct^3$ where a, b and c are constants. When the particle attains zero acceleration, then its velocity will be:

[09 April, 2019]

1. $a + \frac{b^2}{4c}$
2. $a + \frac{b^2}{c}$
3. $a + \frac{b^2}{2c}$
4. $a + \frac{b^2}{3c}$

Question

The displacement 'x' (in meter) of a particle of mass 'm' (in kg) moving in one dimension under the action of a force, is related to time 't' (in sec) by $t = \sqrt{x} + 3$. The displacement of the particle when its velocity is zero, will be **[NEET 2013]**

1. 4 m
2. 0 m (zero)
3. 6 m
4. 2 m

Question

A particle moving along x-axis has acceleration f, at time t, given $f = f_0\left(1 - \frac{t}{T}\right)$, Where f_0 and T are constants. The particle at $t = 0$ has zero velocity. At the instant when $f = 0$, the particle's velocity is **[AIPMT (Prelims)-2007]**

1. $\frac{1}{2} f_0 T$
2. $f_0 T$
3. $\frac{1}{2} f_0 T^2$
4. $f_0 T^2$

Question

A body is moving with variable acceleration (a) along a straight line. The average acceleration of body in time interval t_1 to t_2 is

1. $\frac{a[t_2 + t_1]}{2}$
2. $\frac{a[t_2 - t_1]}{2}$
3. $\frac{\int_{t_1}^{t_2} a\,dt}{t_2 + t_1}$
4. $\frac{\int_{t_1}^{t_2} a\,dt}{t_2 - t_1}$

Question

A particle moves in a straight line and its position x at time t is given by $x^2 = 2 + t$. Its acceleration is given by

1. $\frac{-2}{x^3}$
2. $-\frac{1}{4x^3}$
3. $-\frac{1}{4x^2}$
4. $\frac{1}{x^2}$

Question

A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$. The acceleration of particle is proportional to **[2010]**

1. $(\text{velocity})^{3/2}$
2. $(\text{distance})^{2}$
3. $(\text{distance})^{-2}$
4. $(\text{velocity})^{2/3}$

Question

If acceleration of object $a = 2x^{3/2}$ then find velocity at x where initial at $x = 0$ is 4.

Question

The relation between time t and distance x is $t = \alpha x^2 + \beta x$ where α and β are constants. The retardation is:

1. $2\alpha v^3$
2. $2\beta v^2$
3. $2\alpha\beta v^2$
4. $2\beta^3 v^3$

Question

If $a = 3t^2 + 2t$, initial velocity is 5 m/s. Find the velocity at $t = 4$s. The motion is in straight line, a is acceleration in m/s^2 and t is time in seconds.

Question

The deceleration experienced by a moving motorboat after its engine is shut-off is given by $dv/dt = -kv^3$, where k is a constant. If v_0 is the magnitude of the velocity at shut-off, find the velocity as a function of t.

Question

The motion of a body is given by $dv/dt = 6 - 3v$, where v is in m/s. Find

(a) the velocity in terms of t and

(b) terminal velocity. The motion starts from rest.

NEET - 2024

x (Position) $= at^4 + \beta t^2 + \gamma t + \delta$ find ratio of intial velocity to intial accelcration

(8) NEET - 2006

distance of Particle from origin given by $x = 40 + 12t - t^3$ How long Particle moves before coming to rest.

NEET - 2012

Position $x = 8 + 12t - t^3$ find retardation of Particle when velocity beromes zero.

NEET - 2016

velocity $V = At + Bt^2$ where A and B are constant then distance travelled between 1s to 2sec.

NEET - 2015

velocity $v = \beta x^{-2n}$ then find acceleration as a function of 'x'.

NEET -2005

(Q). The displacement OF Particle varies with time $x = a e^{-\alpha t} + b e^{\beta t}$ then find velocity

JEE - 2023

Acceleration of object $a = 3t + 4$ where intial velocity is zero then velocity at $t = 2$ sec.

NEET - 2025

(Q) Time is given as $t = x^2 + x$ then acceleration of the particle.

$mR^2 Q_2$

(Q) velocity of object $V = a\sin(\omega t)\,\hat{i} + a\cos(\omega t)\,\hat{j}$ then find distance travelled by object in 2sec.

Thank

You